

AI-Powered Project Monitoring Tool

Real-Time Tracking with Predictive Risk and Workload Insights

Rushikesh

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The Problem I

Why Current Tools Fall Short

- Status updates in tools like Jira lag reality until someone manually flags an issue
- Predictive AI features (risk scoring, workload balancing) are usually gated behind expensive enterprise tiers
- Task assignment is still mostly manual guesswork, not based on skill fit or current load
- Small teams and students cannot justify enterprise PPM pricing just for predictive insight

Why This Matters I

Motivation

- Catching risk early (before a status field is updated) prevents last-minute delays
- Reallocating work based on real utilization avoids burnout and idle capacity
- An accessible, AI-driven tool fills a real gap below the enterprise price point
- Building this end-to-end pipeline demonstrates practical AI/data science skills, not just model training

System Architecture I

How the Pieces Fit Together

- Data layer: SQLite database storing projects, tasks, and team members
- AI engine: risk scoring, workload utilization, and skill-based matching, computed directly from the data
- Dashboard: Streamlit app reading and writing the same database in real time
- No separate API layer for this build — the dashboard talks to the database directly, keeping the system simple to build and debug

Core Features I

What the Tool Does

- Real-time dashboards: status breakdown, workload chart, and task table
- AI-driven task analytics: a risk score per task, predicting the likelihood of delay
- AI-driven workload management: utilization tracking with reallocation suggestions
- Skill-based task assignment: ranks team members by skill fit and current load
- Jira-style manual entry form: lets a user add or edit tasks live and see the AI react instantly

Technology Stack I

Tools Used

- Python for all application and data logic
- SQLite as the single source of truth for all data
- Streamlit for the live dashboard and manual entry UI
- pandas for data handling, scikit-learn for the risk-prediction model

Current Progress I

What's Working Today

- Database schema and synthetic dataset generated, including team skills and task requirements
- Live dashboard running: status breakdown, workload chart, and color-coded task table
- Manual entry form working end to end — new or updated tasks appear immediately
- Next: risk-scoring model, reallocation engine, and skill-based assignment ranking

Why This Is Different I

Compared to Jira, Asana, and Similar Tools

- Most existing AI features only read structured fields (dates, % complete), not free-text updates
- Predictive risk and workload balancing are usually locked behind enterprise pricing
- Most tools only show that someone is overloaded — this tool suggests a concrete fix
- Task assignment becomes a ranked, data-backed decision instead of a guess

Summary I

Key Takeaways

- Project monitoring tools exist, but predictive AI and skill-based assignment remain enterprise-only
- This build proves that gap can be closed with a lean, two-day pipeline
- The result is a practical demonstration of end-to-end AI system design, not just a trained model

Future Scope I

Beyond This Build

- Integrate with real Jira/Asana/GitHub data instead of synthetic input
- Add explainability to the risk score so users see why a task was flagged
- Richer resourcing: skill taxonomies, leave calendars, multi-project contention